Shreyas Sunil Gaikwad

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Research Interests

Computational Glaciology	PDE-constrained Inverse Problems	Uncertainty Quantification
Machine Learning	Ice-Ocean Interactions	Data Assimilation
High-Performance Computing	Open Source Software Dev.	Automatic Differentiation
Physical Oceanography	Differentiable Simulations	Generative AI

Education

Ph.D. in Computational Science, Engineering and Mathematics, 2019-2024
The University of Texas at Austin
Thesis: *Bayesian Inversion Over Long Time Scales for the Greenland Ice Sheet*
GPA: 4.0/4.0

M.S. in Computational Science, Engineering and Mathematics, 2019-2021
The University of Texas at Austin
GPA: 4.0/4.0

B.Tech in Mechanical Engineering with Honors, 2015-2019
Minor in Computer Science
Indian Institute of Technology Bombay
Thesis: *Computational Analysis of Subcooled Boiling*
GPA: 9.32/10.0, Department Rank: 4/150

Research and Work Experience

Lead Software R&D Engineer, Cadence Design Systems *2026–Present*
Advisors: Dongxiang Wu, Leo Wang

- Contributing to C++ platform software for a multi-decabillion gate hardware emulation system (Palladium/Protium) used to validate computer chip designs before physical manufacturing
- Diagnosing and resolving urgent customer-reported issues (P0/P1), including subtle edge cases in concurrent, non-deterministic logic, for high-performance, low-latency graph-traversal code underlying real-time hardware emulation
- Pruned bad placements, cutting emulator scheduling trials 50% within 2% of best step count via 3 ML models (pairwise learning-to-rank) on engineered design-scale-invariant features in a low-data regime
- Exploring use of Graph Neural Networks (GNNs) paired with ML to improve Palladium emulation performance
- Developing Google Test-based unit test suite of 200+ cases covering optimization logic and multi-threaded task-bucketing

Graduate Research Assistant, CRIOS, UT Austin 2020-2025
Advisor: Dr. Patrick Heimbach
Ph.D. Thesis: Paleo-reconstruction of Greenland Ice Sheet Dynamics over the Holocene via Bayesian Inversion

- Engineered the Fortran-based ice sheet model SICOPOLIS into a differentiable simulation using open-source Algorithmic Differentiation (AD), enabling scalable Bayesian inversion with efficient adjoint-based gradients
- Solved ill-posed, high-dimensional inverse problems constrained by partial differential equations (PDEs), estimating $\mathcal{O}(10^6)$ uncertain parameters in a continental ice sheet model, accelerated by preconditioned quasi-Newton methods

- Reduced sensitivity analysis cost from $\mathcal{O}(N^3)$ to $\mathcal{O}(1)$ model evaluations using adjoint-based gradients.
- Developed a novel interpretation technique for neural networks using physics-informed adjoints and eXplainable-AI (XAI) methods in Keras, validating learned insights against physically consistent adjoint sensitivities.

Deep Learning Intern, Ansys Summer 2024
Advisor: Jay Pathak, Senior R&D Director, CTO Central Team

- Developed unified Neural Radiance Field (NeRF) to implicitly represent multiple, diverse object geometries.
- Built scalable GenAI pipeline in PyTorch with 106 million parameters using unified NeRF and 3D-aware diffusion models to enable novel 3D object generation from single camera views of object geometries.
- Achieved 3D object generation time of 5 minutes for novel, user-provided single camera-view images.

Visiting Scholar, Argonne National Laboratory Summer 2022
Advisors: Dr. Sri Hari Krishna Narayanan, Dr. Patrick Heimbach, Dr. Michel Schanen

- Open-source software development in Julia, interfacing Enzyme and MPI for a mountain glacier code. Identified fundamental bugs in Enzyme and reported them as Minimum Working Examples (MWEs).
- Developed the first open-source, MPI-parallel, AD-based, adjoint-enabled Bayesian optimization framework for atmospheric and ocean general circulation model MITgcm; alternative to a proprietary tool ($15,000/year/user)

Undergraduate Research Assistant, IIT Bombay 2018-2019
Advisor: Dr. Janani Srree Muralidharan

- Simulated air-water multiphase flow around nuclear rods, analyzing trends in void fraction distribution for different turbulence models and solvers in OpenFOAM.

Undergraduate Research Assistant, IIT Bombay Summer 2017
Advisor: Dr. Shivasubramanian Gopalakrishnan

- Simulated the impact of a subsonic, thin stream of air striking the surface of the water, to mirror the industrial process of purification of molten steel, leveraged axial symmetry to reduce computation time.

Undergraduate Research Assistant, IIT Bombay Summer 2017
Advisor: Dr. S.K. Maiti

- Developed codes in MATLAB to detect the position of transverse cracks with 95% accuracy in Euler-Bernoulli as well as Timoshenko beams using any three modes of natural vibration.

Publications and Talks

Articles

Shreyas Sunil Gaikwad*, Helen Pillar*, Dhruv Apte*, Patrick Heimbach (2026). "Pairing Neural Networks with Adjoints for Flexible Investigation and Robust Attribution of Ocean Variability", *In Preparation for submission to Geophysical Research Letters.* [1]

Fairuz Ishraque, Mohammad Afzal Shadab, Kevin Shionalyn, Neosha Narayanan, Meghan Schaberg, **Shreyas Sunil Gaikwad**, Izuchukwu Ezukanma, Olga Lauter, John A. Higgins (2026), "Assessing Challenges for U.S.-Based Polar Early Career Scientists During Science Policy Changes", *Submitted to Journal of Glaciology, available on Earth-ArXiv.*

Laurent Hascoet, Jean-Luc Bouchot, **Shreyas Gaikwad**, Sri Hari Krishna Narayanan, Jan Hückelheim (2026). "Profiling checkpointing schedules in adjoint ST-AD", *Proceedings of the 2024 International Conference on Algorithmic Differentiation (AD).*

Detlef Stammer, Daniel E. Amrhein, Magdalena Alonso Balmaseda, Laurent Bertino, Massimo Bonavita, Carlo Buontempo, Nico Caltabiano, Francois Counillon, Ian Fenty, Raffaele Ferrari, Yosuke Fujii, **Shreyas Sunil Gaikwad**, Pierre Gentine, Andrew Gettelman, Ganesh Gopalakrishnan, Patrick Heimbach, Hans Hersbach, Chris Hill, Shinya Kobayashi, Armin Köhl, Paul J. Kushner, Matthew Mazloff, Hisashi Nakamura, Stephen G. Penny, Laura Slivinski, Susann Tegtmeier, and Laure Zanna (2024). "Earth system reanalysis in support of climate model improvements", *Bulletin of the American Meteorological Society.*

[1] * *Equal contributions to this manuscript.*

Laurent Hascoet, Jean-Luc Bouchot, **Shreyas Gaikwad**, Sri Hari Krishna Narayanan, Jan Hückelheim (2024). "Profiling checkpointing schedules in adjoint ST-AD", *8th International Conference on Algorithmic Differentiation, 2024.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Jean-Michel Campin, Helen Pillar, An Nguyen, Jan Hückelheim, Paul Hovland, Patrick Heimbach (2024). "MITgcm-AD v2: Open source tangent linear and adjoint modeling framework for the oceans and atmosphere enabled by the Automatic Differentiation tool Tapenade", *Future Generation Computer Systems, 2024.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Liz Curry-Logan, Ralf Greve, Patrick Heimbach (2023). "SICOPOLIS-AD v2: linearized forward and adjoint modeling framework for ice sheet modeling enabled by automatic differentiation tool Tapenade", *Journal of Open Source Software 8, no. 83 (2023): 4679.*

Invited Talks

Fairuz Ishraque, Mohammad Afzal Shadab, Olga Lauter, Neosha Narayanan, Kevin Shionalyn, Meghan N Schaberg, Zachary Labe, **Shreyas Sunil Gaikwad**, Melinda Nicewonger, Rohini Narayanan (2026), "Assessing Challenges for Polar Early Career Scientists During Science Policy Upheaval", *AGU Annual Meeting, 2025.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Liz-Curry Logan, Ralf Greve, Jean-Michel Campin, Helen Pillar, An Nguyen, Jan Hückelheim, Paul Hovland, Patrick Heimbach. "Leveraging Automatic Differentiation (AD) to Improve Ice-Sheet and Ocean Modeling", *ICCOPT VIII, 2025.*

Shreyas Gaikwad. "Basics of Adjoints and Algorithmic Differentiation", *ECCO Summer School, 2025.*

Shreyas Gaikwad. "Basics of Adjoints and Algorithmic Differentiation", *ECCO Hackweek, 2024.*

Shreyas Gaikwad, Helen Pillar, Tim Smith, Nora Loose, Sri Hari Krishna Narayanan, Jean-Michel Campin, Laurent Hascoet, Patrick Heimbach (2023). "Computational Science To Enable Digital Twins Of The Ocean", *6th SIAM Texas-Louisiana Sectional Meeting (SIAM TX-LA), 2023.*

Oral Presentations

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Michel Schanen, Patrick Heimbach. "MITgcm-AD: Open Source Inverse Modeling Framework for the Oceans using the AD tool Tapenade", *ECCO Annual Meeting, 2024.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Michel Schanen, Patrick Heimbach. "MITgcm-AD: Open Source Inverse Modeling Framework for the Oceans using the AD tool Tapenade", *25th EuroAD Workshop, 2023.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Michel Schanen, Patrick Heimbach. "MITgcm-AD: Open Source Inverse Modeling Framework for the Oceans using the AD tool Tapenade", *WCRP Workshop MIT, 2023.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Michel Schanen, Patrick Heimbach. "SICOPOLIS-AD v2: Inverse modeling framework for ice sheet modeling enabled by automatic differentiation", *AGU Annual Meeting, 2022.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Michel Schanen, Patrick Heimbach. "SICOPOLIS-AD v2: Inverse modeling framework for ice sheet modeling enabled by automatic differentiation", *Scientific Committee on Antarctic Research 10th Open Science Conference (SCAR-OSC), 2022.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Liz Curry-Logan, Ralf Greve, Patrick Heimbach. "SICOPOLIS-AD V2: Inverse Modeling Framework for Ice Sheet Modeling enabled by Automatic Differentiation", *Machine Learning for Polar Regions Workshop, 2022.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Liz Curry-Logan, Ralf Greve, Patrick Heimbach. "SICOPOLIS-AD v2: An open-source tangent-linear and adjoint modeling framework for ice-sheet simulation enabled by the AD tool Tapenade", *24th EuroAD Workshop, 2021.*

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Michel Schanen, Patrick Heimbach. "MITgcm-AD: Open Source Inverse Modeling Framework for the Oceans using the AD tool Tapenade", *Summer Argonne Students Symposium (SASSy), 2022.*

Posters

Shreyas Gaikwad, Sri Hari Krishna Narayanan, Laurent Hascoet, Liz Curry-Logan, Ralf Greve, Patrick Heimbach (2022). "SICOPOLIS-AD v2 and MITgcm-AD: linearized forward and adjoint modeling framework for ice sheet and ocean modeling enabled by automatic differentiation tool Tapenade", *14th Joint Laboratory for Extreme Scale Computing (JLESC) Workshop.*

Honors and Awards

- **ACDC+GRISO Summer School**, prestigious climate dynamics summer school in Greenland. 2023
- **Peter O'Donnell Fellow**, Awarded Fellowship worth $24,000 by the Oden Institute. 2019-2020
- **Ph.D. Preliminary Exam**, Cleared exams in applied maths, scientific computing, and maths modeling. 2020
- **AP grade**, in Computer Graphics, awarded rarely for exceptional insights. 2018
- **AP grade**, in Heat Transfer, awarded rarely for exceptional insights. 2017
- **KVPY Fellow**, Conferred by Govt. of India for demonstrating aptitude for research, (Rank - 27/50,000). 2015
- **IISc UG Admission offer**, India's top research institution, offered admission due to KVPY selection. 2015
- **JEE Advanced qualification**, Stage II entrance exam for the IITs, Ranked 509/150,000. 2015
- **JEE Main qualification**, Stage I entrance exam for the IITs, Ranked 983/1.5 million candidates. 2015
- **INChO qualification**, Indian National Chemistry Olympiad, ranked top 1% among 30,000 students. 2015
- **INSPIRE Scholar**, Scholarship for top 1 percentile in state high school exams. 2015
- **BITSAT Qualification**, Entrance exam for the BITs, estimated rank top 200/300,000. 2015
- **Bhaskar Genius Scholar**, Top 20 rank in India among 50,000 students. 2012
- **MCSE PSS Scholar**, Awarded by Maharashtra State Council of Examination (99.9995 percentile). 2011

Other Research Projects

Machine Learning Applications in Geophysics Spring 2021

- Developed CNNs in **Keras** for earthquake detection using data from stations, with **96% validation accuracy**.
- Developed U-Nets in **Keras** for seismic faults detection through image segmentation, with **97% test accuracy**.
- Leveraged Autoencoders in **Keras** to aid clustering through dimensional reduction into the latent space.
- Built pipeline to pick mudrocks from real wireline logs using ML algorithms, with **87% test accuracy**.

Physics-Informed Machine Learning Spring 2021

- Trained a Deep Neural Network in **PyTorch** to emulate a partial differential equation (PDE) based glacier model by leveraging **higher-order derivatives of the PyTorch computational graph**.
- Enriched the framework using an expanded computational graph to infer unknown PDE parameters.

Laplacian 2D Finite Difference (FD) Solver Application Fall 2020

- Features: OOP (C++), Solver (gauss, jacobi, PETSc), tests (bats, Travis CI, docker), 98% code coverage (lcov), 0% memory errors (valgrind), build (autotools), HPC env (SLURM), parser & logger (GRVY).

Software contributions

All of these tools are constantly under development.

- **diffblas**, a package that provides open-source differentiated BLAS routines from their implementation in LAPACK using the open-source Automatic Differentiation (AD) tool Tapenade.
- **SICOPOLIS-AD v2**, a new inverse modeling framework for the ice sheet model SICOPOLIS that is enabled by source transformation using the open-source Automatic Differentiation (AD) tool Tapenade.

- **MITgcm-AD v2**, a new inverse modeling framework for the general circulation model MITgcm that is enabled by source transformation using the open-source Automatic Differentiation (AD) tool Tapenade.
- **ARGOVIS**, developed OOP-based Python tools for interactive plotting and binning of Argo data.
- **Enzyme.jl**, Identified fundamental bugs in Enzyme.jl, reported as MWEs here and here by Dr. Michel Schanen (my co-advisor at Argonne).

Select graduate coursework

Data Science and HPC - Machine Learning applications in Geophysics, Foundational Techniques in Machine Learning, Engineering Data Mining, Design and Analysis of Algorithms, Tools and Techniques in Computational Science, Numerical Methods for Differential Equations, Numerical Linear Algebra, High Performance Scientific Computing, Operating Systems.

Inverse Problems and UQ - Uncertainty Quantification in Computational Models, Computational & Variational Methods for Inverse Problems (observer).

Mathematics - Functional Analysis, Partial Differential Equations, Analytical Methods.

Fluid Dynamics - Physical Oceanography, Mathematical Modeling in Science and Engineering, Computational Fluid Dynamics, Galerkin Methods.

Computational Skills

Languages - Python (scikit-learn, Keras, PyTorch, FENICS, xarrays, numpy, autograd, Pandas), Julia (MPI, Enzyme), C++ (GRVY, MASA, HDF5, PETSc), Fortran-77/90.

Softwares - MATLAB, Scilab, Solidworks, ANSYS, ADAM, OpenFOAM, Simulink, LaTeX, Tapenade, MITgcm, SICOPOLIS.

HPC and other tools - OpenMP, MPI, CUDA, SLURM, Github, Docker, Shell scripting, Travis-CI, Autotools, Valgrind, HTML, CSS, Jekyll, lcov.

Summer Schools

Advanced Climate Dynamics Courses (ACDC) + Greenland Ice Sheet Ocean (GRISO) Summer School, Disko Island, Greenland, 2023.

Trustworthy Artificial Intelligence for Environmental Science (TAI4ES) Summer School, 2022.

Trustworthy Artificial Intelligence for Environmental Science (TAI4ES) Summer School, 2021.

Professional Memberships

Society of Industrial and Applied Mathematics (SIAM)

International Association of Cryospheric Sciences (IACS)

American Geophysical Union (AGU)

Indian Society of Arctic Researchers (ISAR)

Arctic Research Consortium of the United States (ARCUS)

United States Association of Polar Early Career Scientists (USAPECS)

Seismological Society of America (SSA)

Volunteer, Mentorship, Leadership, Teaching and Community Experience

Volunteer Experience

Volunteer: CURED, Techfest-2016, IIT Bombay, Conducted 178 diabetes screening camps across 20 cities in a team of 100+, conducting more than 65,000 screenings in a day as part of CURED (Can You Really Escape Diabetes?), a PAN-India level initiative to raise awareness about diabetes.

Volunteer: Educational Outreach, National Service Scheme, IIT Bombay, Mentored and taught 25 underprivileged students of secondary school over a period of one year (2015-2016) as part of the Educational Outreach program of National Service Scheme.

Volunteer: UT Girl Day 2024, Going to help organize and demonstrate to 500+ elementary and middle school kids and their parents how to conduct simple yet powerful experiments to understand the role of sea ice and glacial melts in the changing Arctic climate and explore observational and modeling tools fundamental to many CRIOS research projects.

Mentorship Experience

Mentor: Oden Institute for Computational Engineering and Sciences 2022-2023

- Mentored a first-year Oden Institute graduate student (Bugra Yalcin), served as a contact point for becoming acclimated to the program.

Mentor: UT Austin SIAM Applied Math Mentorship Program 2021-2022

- Mentored 2 students (Martin Vu and Nan Shelden), with broad focus on careers in applied mathematics.

Mentor: Department Academic Mentorship Program (DAMP), IIT Bombay 2018-2019

One of the 23 student mentors selected out of 85 applicants, selections based on interviews and peer reviews.

- Mentored 6 students for a smooth transition to department curriculum, motivating their academic and personal endeavors (Students mentored - Rishabh D'Souza, Ameya Mittal, Pranesh Parmar, Nishant Kumar, Harshit Madan, Swastik Sahoo).
- Part of the team tasked with suggesting modifications to the curriculum if necessary after consultation with the HoD, Faculty Advisors, and Professors.

Mentor: Academic Rehabilitation Program, IIT Bombay 2018-2019

- Counselled an academically weak student on a one-on-one basis, helping him to clear his academic backlogs as part of the Academic Rehabilitation Program.

Informal career mentorship 2018-Present

- Mentored over 20+ students broadly about future careers in STEM, interview preparation, and resume reviews.

Volunteering and Leadership Experience

Board member and website lead, USAPECS 2024-Present

- USAPECS is the national committee representing United States members of the Association of Polar Early Career Scientists (APECS). Our aim as USAPECS is to stimulate inclusive collaborations and connections between early career researchers and polar organizations.

Senior Advisor and President, Austin Chapter of SIAM 2022-2024

- Hosted NREL, Sandia National Laboratories, AMD as part of our Industry Seminar Series.
- Helped organize Applied Maths Mentorship Program, which comprised 40 mentors and 50 mentees in total.
- Helped fund and mentor 3 teams of undergraduates for the COMAP Mathematical Contest in Modeling (MCM) / Interdisciplinary Contest in Modeling (ICM).
- Initiated the Recent Graduate Series of talks by Oden Institute Alumni.

Vice President, Austin Chapter of SIAM 2021-2022

- Hosted ANSYS, Two Sigma, Julia Computing, StriveWorks, Cerfe Labs (ARM), and Sandia National Laboratories as part of our Industry Seminar Series.
- Hosted Dr. Howard Stone (Donald R. Dixon '69 and Elizabeth W. Dixon Professor and the Department Chair of the Mechanical and Aerospace Engineering at Princeton University) for a talk on "Beauty in Research and Intersections with Teaching" as part of our Distinguished Seminar Series.
- Quadrupled our followers on Instagram and increased our Facebook following by 30%.
- Helped organize Applied Maths Mentorship Program, which comprised 34 mentors and 47 mentees in total.
- Nominated as finalist for Best Graduate Organization category of Swing Out Awards, for the second year in a row.

Teaching Experience

Graduate Teaching Assistant, UT Austin Fall 2022
Course *- Intro. to Mathematical Modeling in Science and Engineering I,* ***Advisor*** *- Dr. Patrick Heimbach*

- Responsible for grading assignments, conducting recitation and office hours, as well as full lectures in the absence of the instructor.

Community Participation

Lead instructor, ECCO Hackweek 2024

I was one of the lead instructors for the ECCO Hackweek 2024. My responsibilities included giving lectures and developing tutorials and group projects on the use of algorithmic differentiation to generate adjoint models of the ocean model, MITgcm.

Full course lecture, Introduction to Mathematical Modeling in Science and Engineering I 2022

Instructor: Patrick Heimbach

My lecture covered the material around Lagrangian and Eulerian derivatives, as well as the proof of the Reynolds' Transport Theorem.

ARGOVIS Hackathon 2022

I helped develop OOP-based Python tools for interactive plotting and binning of Argo data. The Jupyter Notebook will soon be available and help the community to better analyze and visualize the Argo data.

Guest Lecture, Physical Oceanography 2021

Instructor: Patrick Heimbach

My guest lecture "Scaling Analysis" focused on 1) an overview of how scaling analysis is performed and 2) on how it can help greatly simplify the governing equations of ocean flow by eliminating most of the terms.

Other Industry Experience

KITES Intern, ITC Limited Summer 2018
ITC Limited is India's largest FMCG conglomerate; I interned in the packaging section of its only fully owned flour plant in Uluberia, West Bengal, India
Advisor: Aurko Chatterjee

- Achieved Manpower Reduction in Primary Packing Machines by 50%.
- Secondary Packing Machines worth a total of $165,000 made operational using PDCA-cycle-based improvements.

Junior Design Engineer, IIT Bombay Racing 2016-2017
A team of students which fabricates an electric race car for Formula Student U.K

- Developed 3-D CAD models for gearbox in SolidWorks and carried out FEA simulations in ANSYS to help select optimum gear ratio.
- Routed the cooling system on SolidWorks Routing, reducing weight and increasing ease of priming of the pumps.

References

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